# Find Your Way Around

## THE FREEDAWM GUIDE



ONE PAGE, ONE IDEA:
CHEAT SHEETS FOR RECORDING, EDITING AND MIXING

FreeDAWm

# Welcome to FreeDAWm

## A free, open-source studio for recording, editing and mixing music, on the computer you already have.

FreeDAWm is a digital audio workstation. It runs on macOS, Windows, Linux and FreeBSD, and in a web browser.

You work in two main windows. The **Edit** window is the timeline, where you record, cut and arrange. The **Mix** window is the mixer, where you balance and shape the sound.

This guide is a binder of cheat sheets, not a manual to read from start to finish. Each page covers one idea, and most are a picture with labels.

Keep the page you need open beside FreeDAWm, or print it and pin it up.

### HOW TO READ A SHEET

- The lettered dots on a picture match the lettered notes under it.
- Keys look like this: `Cmd` `S`. Hold the first, press the second.
- On Windows, Linux and FreeBSD, press `Ctrl` wherever a sheet says `Cmd`.
- Menu items are written as a path: Setup › Keyboard Shortcuts.

The pictures show **Midnight Groove**, the demo song that comes with FreeDAWm, so your screen can match the page while you follow along.

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SECTION 1

# Getting Around

The two windows you will spend your time in, the four ways clips can move, and the keys worth learning first.

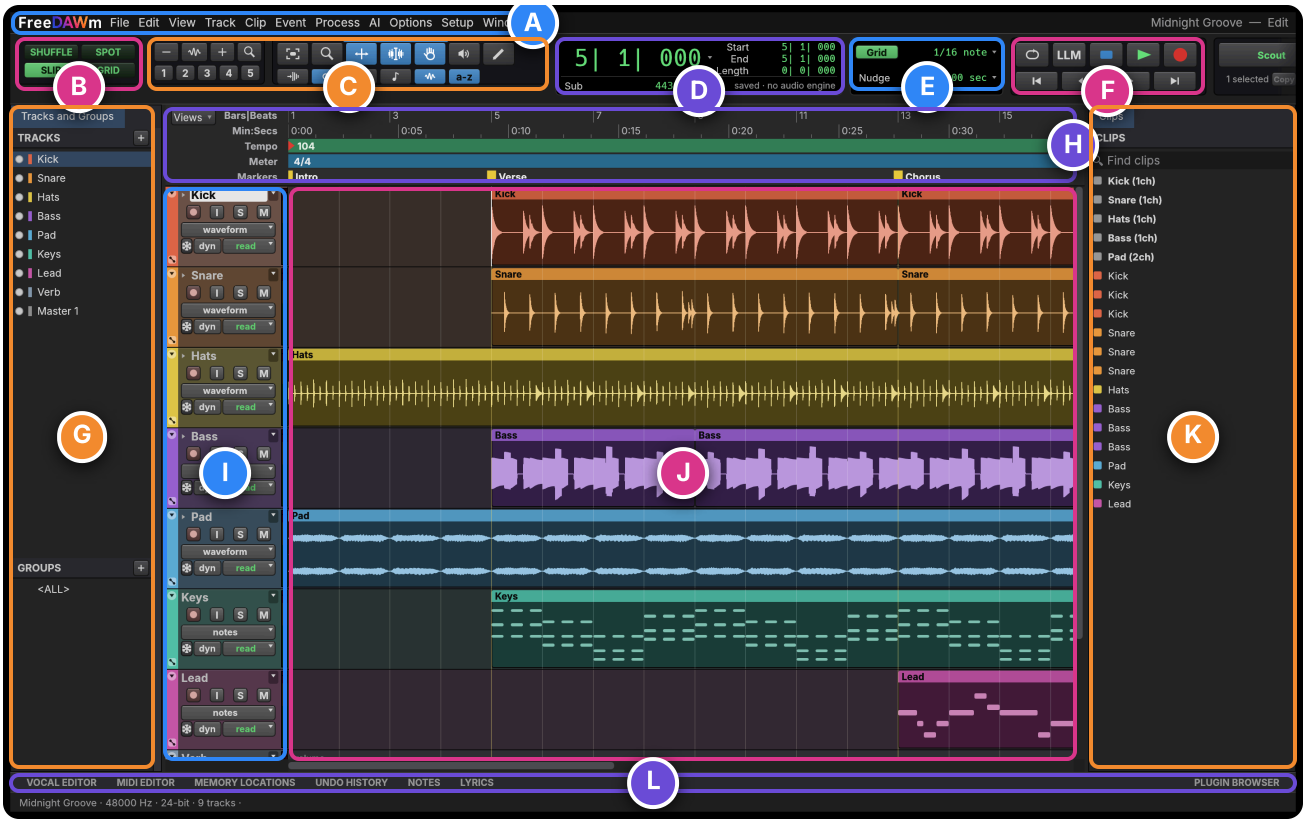
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# THE EDIT WINDOW

**The Goal**

Know what each part of the timeline window is for, so you can find a control without hunting. Open it with Window › Edit, or press Cmd = to flip between Edit and Mix.




**A — Menu bar**
Every command lives here. If you forget a key, look in the menus.

**B — Edit modes**
Shuffle, Slip, Spot and Grid decide how clips move when you drag them. See page 7.

**C — Zoom and tools**
Zoom buttons and five zoom presets, then the editing tools. See page 7.

**D — Counter**
Where the cursor is, and the start, end and length of your selection.

**E — Grid and Nudge**
The size of a grid step and of a nudge. Nudge with + and −.

**F — Transport**
Loop, stop, play and record on top, and the buttons that jump or wind underneath.

**G — Tracks and Groups**
Every track in the song. Click the dot beside a name to show or hide that track.

**H — Rulers**
Time in bars and in minutes, then tempo, meter and markers.

**I — Track controls**
Record, input monitor (I), solo (S) and mute (M), then the track's view and automation mode.

**J — Timeline**
Your clips. Audio is drawn as waveforms and MIDI as notes. Time runs left to right.

**K — Clips list**
Every clip in the session. Type in Find clips to search it.

**L — Panel tabs**
Click a tab to open the Vocal Editor, MIDI Editor, Undo History and more.

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# THE MIX WINDOW

**The Goal**

Read a channel strip from top to bottom. Sound enters at the top of a strip and leaves at the fader, so the order on screen is the order the sound travels.




**A** **Transport and counter**
The same play, stop and record controls as the Edit window.

**B** **Channel width**
Slim strips fit more on screen. Wide strips show each effect's knobs.

**C** **Effects**
EQ, compressors and the rest, processed from top to bottom. Click + to add one.

**D** **Sends**
Slots a to e each send a copy of the track somewhere else, often to a shared reverb.

**E** **I/O**
Where the track's sound comes from and goes to, then its automation mode.

**F** **Group and pan**
The knob places the sound left or right. Stereo tracks get one knob per side.

**G** **Strip buttons**
Input monitor, record and freeze, then solo (S) and mute (M).

**H** **Fader and meter**
The fader sets the track's level. The meter beside it shows how loud it is.

**I** **Level and name**
The fader's value in dB, the track type, and the track name.

**J** **Aux input**
Verb is an aux track. It takes whatever the other strips send to Verb.

**Good to know**

The list on the left is the same Tracks and Groups list as in the Edit window. The coloured bar under each name matches that track's clips on the timeline.

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# EDIT MODES & TOOLS

**The Goal**

The edit mode decides where a clip lands when you move it. The tool decides what a click or drag does. Pick both before you edit.

## THE FOUR EDIT MODES

BY HAND

### Slip
F2 or Alt+2



Clips go exactly where you drop them. Gaps and overlaps are allowed. This is the mode FreeDAWm starts in.

### Shuffle
F1 or Alt+1



Clips close up end to end. Cut a piece out and everything after it slides left to fill the gap.

### Spot
F3 or Alt+3




FreeDAWm asks for an exact position, and the clip goes there. Use it when you know the time and not the place.

### Grid
F4 or Alt+4



Clips and selections jump to the nearest grid line. Press F4 again for Relative Grid, which keeps a clip's offset from the line.

STAYS WHERE YOU PUT IT

SNAPS INTO PLACE

BY THE CLOCK

**Left and right:** does the clip stay where you put it, or does FreeDAWm line it up for you?

**Up and down:** do you place it by hand, or by a time on the ruler?

## THE TOOLS

- F5 **Zoomer** — Drag across an area to zoom in on it.
- F6 **Trim** — Drag a clip's edge to make it shorter or longer.
- F7 **Selector** — Click to place the cursor. Drag to select a range.
- F8 **Grabber** — Drag a clip to move it.
- F9 **Scrubber** — Drag over audio to hear it under the pointer.
- F10 **Pencil** — Draw notes and automation.

**Smart Tool**

Press F6 and F7 together for three tools in one. Near a clip's edge it trims, in the upper half of a clip it selects, and in the lower half it grabs.

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# KEYBOARD SHORTCUTS

**The Goal**

Learn these first. They cover most of a session, and each one saves a trip to the menus. On Windows, Linux and FreeBSD, press Ctrl wherever you see Cmd.

## PLAY AND RECORD

| | |
|---|---|
| Play or stop | Space |
| Record | F12 or Cmd Space |
| Play at half speed | Shift Space |
| Back to the start | Home |
| Go to the end | End |
| Add a marker | Enter |
| Loop playback on or off | Cmd Shift L |
| Play the selection | Alt [ |

## TRACKS

| | |
|---|---|
| New track | Cmd Shift N |
| Record enable | Shift R |
| Solo | Shift S |
| Mute | Shift M |
| Input monitoring | Shift I |
| Group tracks | Cmd G |

## SESSION

| | |
|---|---|
| New session | Cmd N |
| Open a session | Cmd O |
| Save | Cmd S |
| Save as | Cmd Shift S |
| Import audio | Cmd Shift I |
| Bounce the mix to a file | Cmd Alt B |

## EDITING

| | |
|---|---|
| Undo | Cmd Z |
| Redo | Cmd Shift Z |
| Cut, copy, paste | Cmd X C V |
| Duplicate | Cmd D |
| Select all | Cmd A |
| Split at the selection | Cmd E |
| Heal a split | Cmd H |
| Trim to the selection | Cmd T |
| Create fades | Cmd F |
| Mute a clip | Cmd M |
| Clear the selection | Cmd B or Delete |
| Nudge later or earlier | + − |

## ZOOM AND VIEW

| | |
|---|---|
| Zoom out or in | R T |
| Fit the whole session | Alt A |
| Zoom to the selection | Alt F |
| Zoom toggle | E |
| Jump to the next clip edge | Tab |
| Switch between Mix and Edit | Cmd = |
| Edit modes | F1 to F4 |
| Tools | F5 to F10 |

**Make them yours**

Change any shortcut in Setup › Keyboard Shortcuts. A few keys are fixed, including Space, Enter, Tab and the F keys.

SECTION 2

# Recording

Get sound into FreeDAWm: choose your audio device, make a track, record a take, and fix the part that went wrong without redoing the rest.

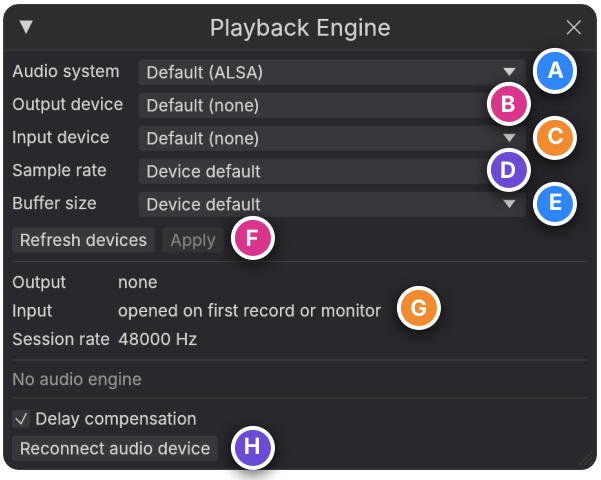
FreeDAWm

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# SET UP YOUR SOUND

**The Goal**

Tell FreeDAWm which device to play through and record from. You do this once per device, in Setup › Playback Engine.




**A — Audio system**
How FreeDAWm talks to your sound hardware. Leave it on Default unless your interface needs another.

**B — Output device**
Where you hear the mix: your interface, speakers or headphones.

**C — Input device**
Where recordings come from: your interface or a microphone.

**D — Sample rate**
The detail of the sound. Device default is a safe choice.

**E — Buffer size**
How much sound is handled at a time. See below.

**F — Refresh and Apply**
Refresh after you plug something in. Apply makes your choices take effect.

**G — Status**
What is open right now, and the session's sample rate.

**H — Reconnect**
Click this if the sound stops after a device was unplugged.

## BUFFER SIZE: A TRADE

SMALL — LARGE

### Small buffer

You hear yourself with almost no delay, but the computer works harder and may crackle. Best while recording.

### Large buffer

Steady playback with lots of effects, but a delay you can hear when you play or sing. Best while mixing.

**Good to know**

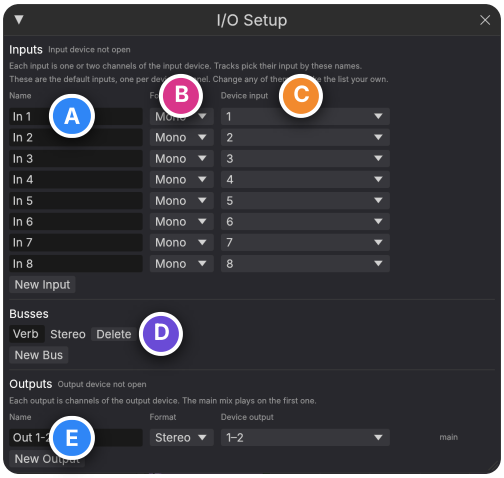
On the picture the input reads "opened on first record or monitor". FreeDAWm only opens your microphone when you arm a track or turn on monitoring.

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# INPUTS & NEW TRACKS

**The Goal**

Give your inputs names you will recognise, then make a track to record on. Named inputs save you from guessing which socket is which.




**A** **Input names**
Open Setup › I/O and type over a name: Vocal Mic, Guitar DI. Tracks pick their input by these names.

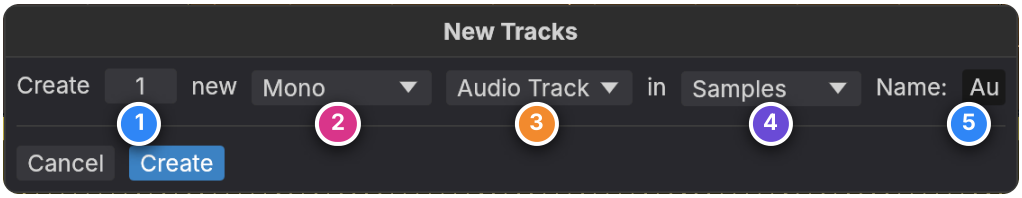
**B** **Format**
Mono is one channel, for a single mic or cable. Stereo is a pair.

**C** **Device input**
Which socket on your interface this name stands for.

**D** **Busses**
Paths inside FreeDAWm, used to send several tracks to one place.

**E** **Outputs**
Where sound leaves. The main mix plays on the first one.

## MAKE A TRACK




Track › New, or Cmd Shift N

**1** **How many**
Make several tracks in one go.

**2** **Mono or stereo**
Match the input you will record.

**3** **Track type**
See the table below.

**4** **Samples or ticks**
Ticks make clips follow tempo changes.

**5** **Name**
Name it now. Recordings take the track's name.

| | |
|---|---|
| **Audio Track** | Records and plays sound: a voice, a guitar, a drum mic. |
| **Instrument Track** | Plays a software instrument from the notes you record or draw. |
| **MIDI Track** | Holds notes only, to drive an instrument somewhere else. |
| **Aux Input** | Has no clips. It takes sound from a bus, often for a shared reverb. |
| **Master Fader** | Controls the level of an output, usually the whole mix. |

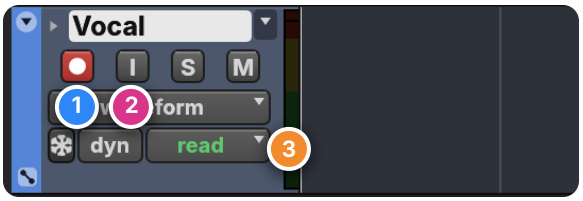
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# RECORD A TAKE

**The Goal**

Record your first take in six steps. The first three happen on the track, the last three on the transport.

## ON THE TRACK

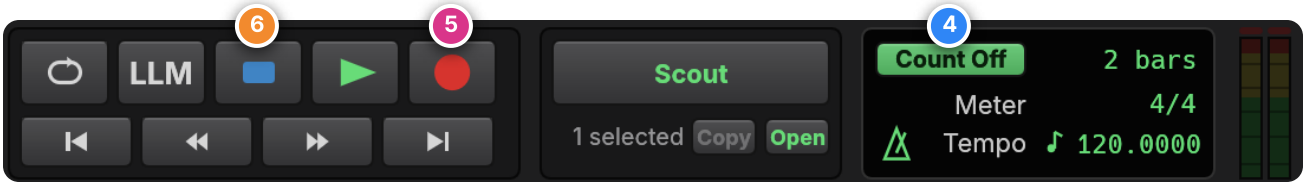



1. **Arm the track**
   Click its record button so it lights red, or select the track and press Shift R.

2. **Hear yourself**
   Click I to monitor the input through the track, effects included. Shift I does the same.

3. **Check the level**
   Play or sing your loudest part and watch the meter. Turn the input down on your interface if it reaches the red at the top.

## ON THE TRANSPORT




4. **Count yourself in**
   Click Count Off to hear two bars of click before the take starts.

5. **Record**
   Click the red button, or press F12 or Cmd Space.

6. **Stop**
   Press Space. Your take appears as a clip on the track.

## ALSO ON THE TRANSPORT

| | |
|---|---|
| **Metronome** | The icon beside Tempo turns the click on or off during the take. |
| **LLM** | Low Latency Monitoring. It bypasses the effects that add delay on the tracks you are recording. |
| **Loop** | Plays the selection round and round. See the next page for recording in a loop. |
| **Tempo and Meter** | Set these before you record to a click. Drag the tempo, or click it to type a number. |

**Not happy?**

Press Cmd Z to undo the take, then record again. FreeDAWm autosaves as you work and can recover a session after a crash.

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# PUNCH, LOOP & COMP

**The Goal**

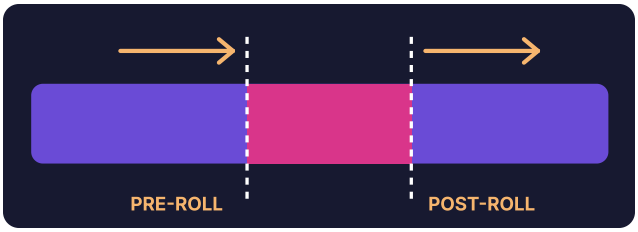
Fix one line without redoing the song, record several tries in a row, and build one good take from the best parts of each.




## Punch in

Select the part you want to redo, then record. Only the selection is replaced. Everything around it stays.

No selection? Press Record while the song plays to drop in at that moment.




## Pre-roll and post-roll

Turn on Options › Pre/Post-Roll, or press Cmd K. Playback starts a little before the punch so you can play along into it, and carries on a little after.




## Loop record

Turn on Options › Loop Record, or press Alt L. Select the bars to repeat and record. Each time round makes a new take, so you can keep trying without stopping.




## Comp the best parts

Click the small arrow before the track name to show its take layers. Drag across the part of a take you like and it is copied up into the track, with a short crossfade at each join.

SECTION 3

# Editing

Shape what you recorded: cut and move clips, smooth the joins with fades, fix notes in the MIDI Editor, and find your way round a long song.

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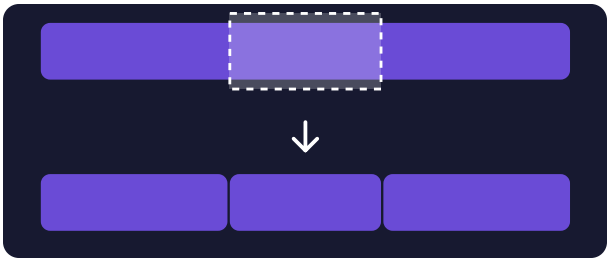
# CUT, MOVE & TIDY

**The Goal**

Six edits do most of the work. Each one acts on the selection, so select first: drag across a clip with the Selector, or click once to place the cursor.

## Split

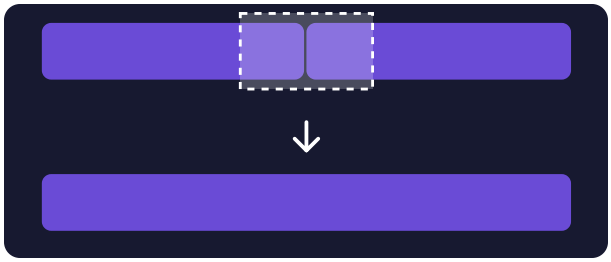
Cmd E



Cuts the clip at both edges of the selection. With only a cursor, it cuts once.

## Heal

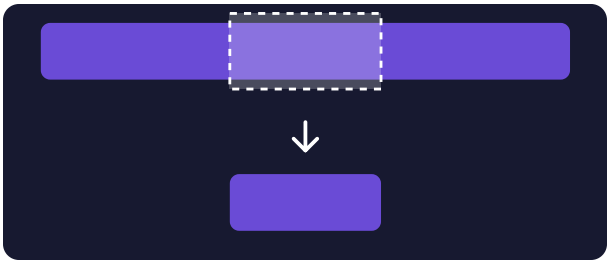
Cmd H



Select across a cut and heal it. The two pieces become one clip again.

## Trim to selection

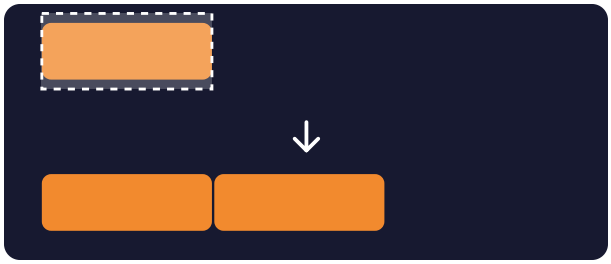
Cmd T



Keeps the selected part of the clip and removes the rest.

## Duplicate

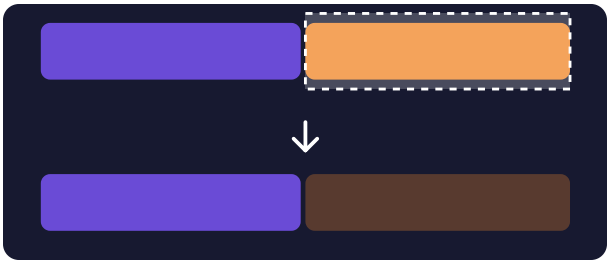
Cmd D



Places a copy straight after the selection. Press it again for another.

## Mute

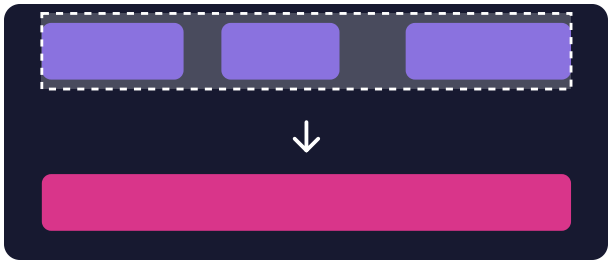
Cmd M



The clip stays where it is but makes no sound. Press again to bring it back.

## Consolidate

Alt Shift 3



Turns the selection, gaps included, into one new clip.

**Good to know**

These edits change clips, not your recordings. To get trimmed sound back, drag the clip's edge out again with the Trim tool. Cmd Z undoes any of them.

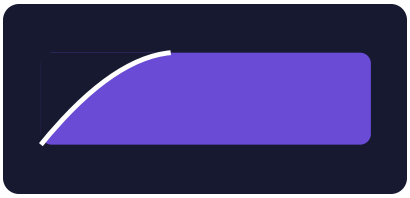
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# FADES & CROSSFADES

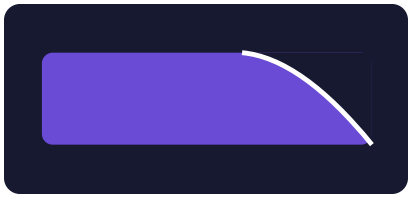
**The Goal**

A cut that lands in the middle of a sound can click. A fade ramps the level so the join is smooth. Put one on every edit you can hear.

## THREE KINDS



**Fade in**
Select from before a clip's start to where the sound should be at full level.



**Fade out**
Select from where the fade should begin to past the clip's end.

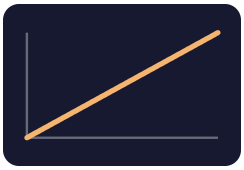


**Crossfade**
Select across the join between two clips. One fades out as the other fades in.

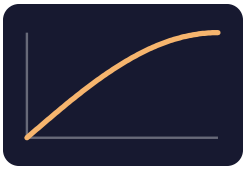
## MAKE ONE

1. **Select**
   Drag across the start, the end or the join, as shown above.
2. **Create**
   Press Cmd F, or choose Edit › Fades › Create.
3. **Pick a shape**
   Choose one of the five below and click OK.

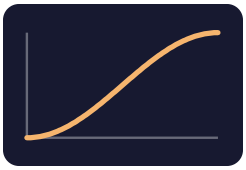
## FIVE SHAPES



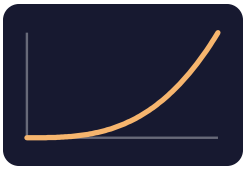
**Linear**
A straight ramp.



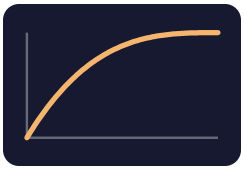
**Equal Power**
Keeps a crossfade's level steady. The usual choice.



**S-Curve**
Gentle at both ends.



**Exponential**
Starts slowly, then rises fast.



**Logarithmic**
Rises fast, then eases in.

Each curve shows the level of a fade in, from silence on the left to full on the right. A fade out is the mirror image.

**Faster**

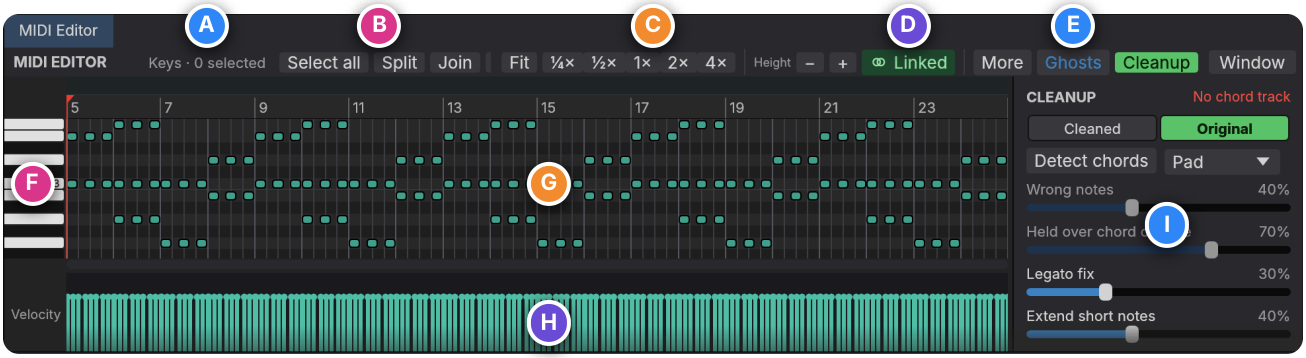
With the Smart Tool, drag from the top corner of a clip to draw a fade by hand. Edit › Fades › Delete removes the fades in the selection.

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# THE MIDI EDITOR

**The Goal**

Fix and write notes. Select a MIDI clip and choose Window › MIDI Editor, or click the MIDI Editor tab at the bottom of the Edit window.




**A — Clip and selection**
The clip you are editing and how many notes are selected.

**B — Note tools**
Select all, Split at the playhead, and Join.

**C — Zoom and height**
Fit shows the whole clip. ¼× to 4× zoom in time. Height makes the rows taller.

**D — Linked**
Keeps the editor lined up with the Edit window's timeline.

**E — Ghosts**
Shows the notes of other MIDI clips faintly, so you can write against them.

**F — Piano keys**
Each row is one pitch. Click a key to hear it.

**G — Notes**
Each bar is a note. Higher is higher in pitch, longer lasts longer. Drag to move one.

**H — Velocity**
How hard each note is played. A taller bar is a louder note.

**I — Cleanup**
Tidies a played part. Compare Cleaned with Original, then Commit.

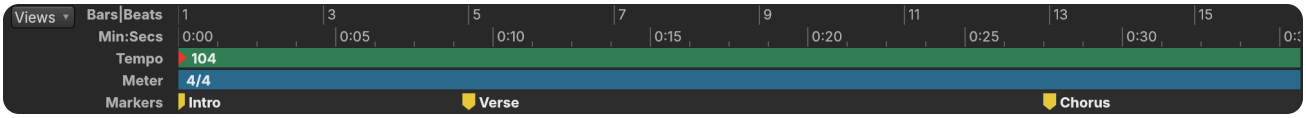
| | |
|---|---|
| **Play the selection** | `Space` with the pointer over the notes |
| **Move without snapping** | Hold `Alt` while you drag |
| **Cancel a drag** | `Esc` |
| **Fit the clip** | `F` with the pointer over the notes |
| **Quantize, Transpose** | `Alt` `0`, `Alt` `T` (also in the Event menu) |

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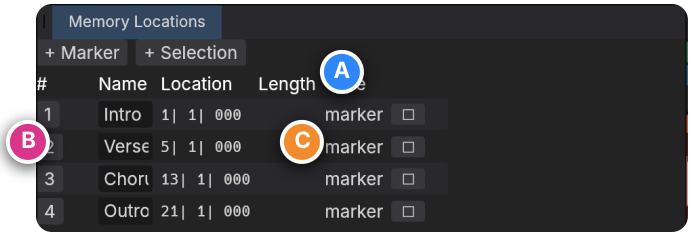
# MARKERS & MOVING AROUND

**The Goal**

Label the parts of your song once, then jump between them. A few keys cover the rest: zooming, stepping from clip to clip, and nudging.




Markers show as flags on the Markers ruler, the bottom row here.




**A** **+ Marker and + Selection**
Marker stores the cursor's place. Selection stores the selected range. Enter also adds a marker.

**B** **Number**
Type . then the number then . to jump to that location.

**C** **Name and location**
What you called it and where it sits in the song.

Window › Memory Locations, or the Memory Locations tab at the bottom of the Edit window

## ZOOM

| | |
|---|---|
| Zoom out or in | R T |
| Fit the whole session | Alt A |
| Zoom to the selection | Alt F |
| Zoom toggle | E |
| Taller waveforms | Cmd Alt ] |
| Recall a zoom preset | Click 1 to 5 |
| Store a zoom preset | Cmd-click 1 to 5 |

## MOVE AND NUDGE

| | |
|---|---|
| Start or end of the session | Home End |
| Next clip edge | Tab |
| Previous clip edge | Alt Tab |
| Extend the selection as you go | Add Shift |
| Nudge later or earlier | + − |
| Bigger or smaller grid | Shift = − |
| Bigger or smaller nudge | Alt Shift = − |

**Tab to Transient**

Turn on Options › Tab to Transient and Tab stops at each hit inside a clip, not only at its edges. It is the quick way to find a drum hit to cut on.

SECTION 4

# Mixing

Balance the tracks, shape each sound with effects, share one reverb between many tracks, and make the mix move with automation.

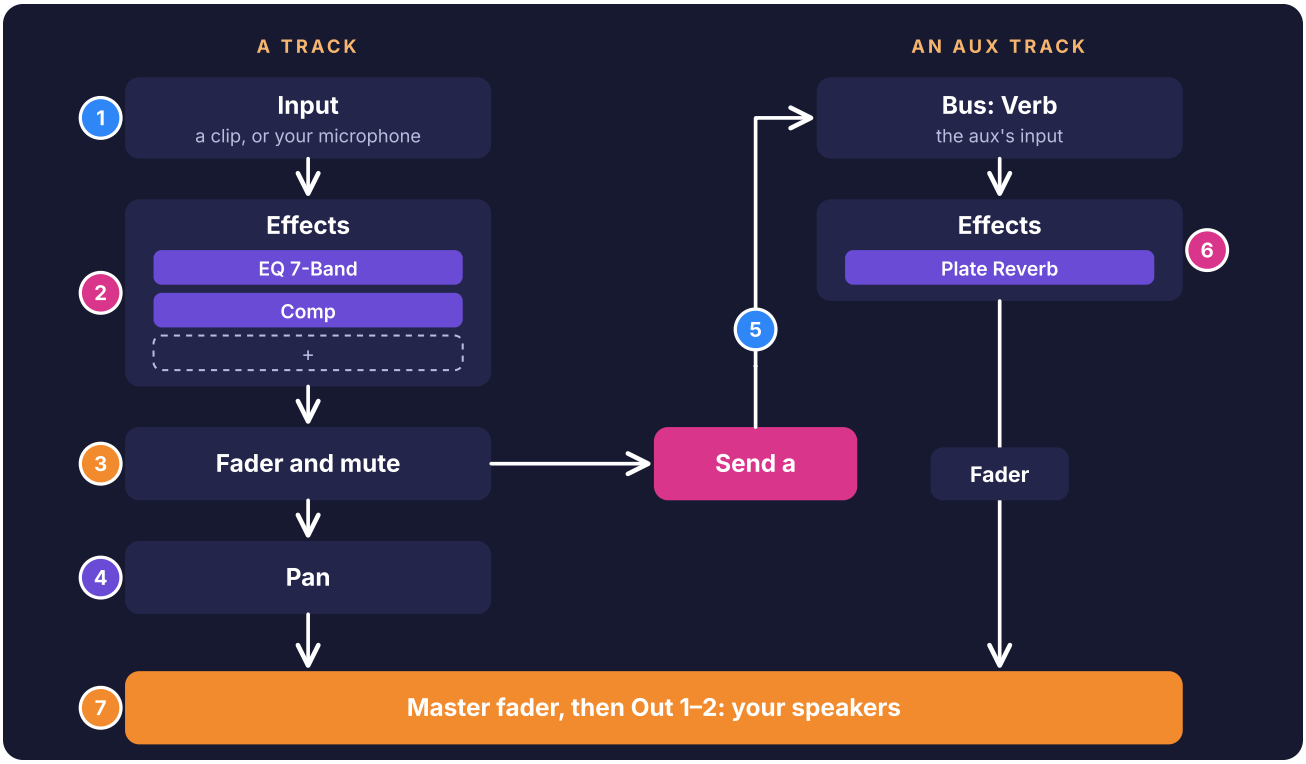
FreeDAWm

FreeDAWm

# SIGNAL FLOW

**The Goal**

Know the path sound takes through the mixer. Once you can trace it, you can work out why a change did or did not do what you expected.




1. **Input**
   What the track plays: its clips, or the live input while you record.
2. **Effects**
   Processed in order from the top. Swap two and the sound changes.
3. **Fader and mute**
   Sets the track's level after the effects have done their work.
4. **Pan**
   Places the track between the left and right speakers.
5. **Send**
   A copy of the track, taken after the fader, on its way to a bus.
6. **Aux track**
   Listens to the bus and adds its own effects, here one reverb for every track that sends to it.
7. **Master**
   Every track and aux ends up here. Its fader sets the level of the whole mix.

**Try this**

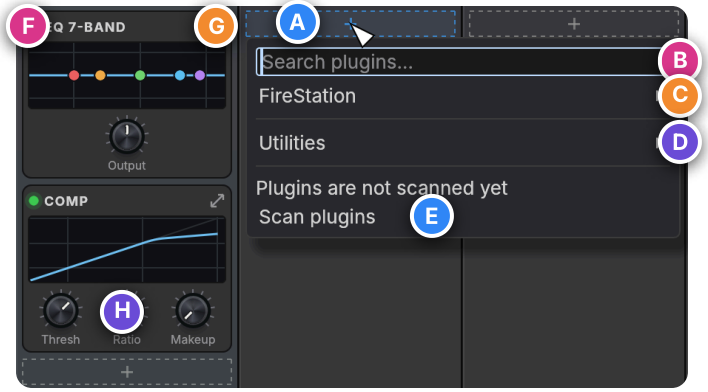
A compressor sits before the fader, so turning the fader down does not change how hard it works. To drive it harder or softer, change the level before it: the clip's gain or the effect's own input.

FreeDAWm

# ADD AN EFFECT

**The Goal**

Put an effect on a track and adjust it. In the Mix window each effect is a tile at the top of the strip, and an empty slot is a dashed box with a plus.




**A — Empty slot**
Click the plus to open the list of effects.

**B — Search**
Type part of a name, such as comp or verb.

**C — FireStation**
The built-in studio effects: compressor, EQ, tape, reverb and more.

**D — Utilities**
The simpler built-in tools.

**E — Scan plugins**
Finds the plugins already installed on your computer and adds them to this list.

**F — On light**
Green means the effect is working.

**G — Open**
The arrow opens the effect's full editor in its own window.

**H — Quick knobs**
The controls you reach for most, on the strip itself.

## THREE HABITS

1. **Listen in the mix**
Adjust an effect with the whole song playing, not with the track soloed. What matters is how it sits with the rest.

2. **Match the level**
Louder nearly always sounds better. Compare with the effect off at the same loudness before you decide.

3. **Mind the order**
Effects run top to bottom. An EQ before a compressor changes what the compressor reacts to.

**Running slow?**

The snowflake on a strip freezes the track. FreeDAWm plays a rendered copy and the effects stop using the computer. Click it again to unfreeze and carry on editing.

FreeDAWm

# WHAT TO REACH FOR

**The Goal**

Match the problem you hear to the kind of effect that fixes it. Every effect named here comes with FreeDAWm.

| THE JOB | WHAT YOU HEAR, AND WHAT THE EFFECT DOES | REACH FOR |
|---|---|---|
| **Tone** | Muddy, harsh, thin or dull. An EQ turns ranges of the sound up or down, from the low rumble to the airy top. | **EQ 7-Band, EQ 1-Band, CleanQ, Filter** |
| **Level** | Some notes jump out and others get lost. A compressor turns the loud moments down so the part sits steady. | **Compressor/Limiter, Comp, Channel Strip** |
| **Noise** | Hiss or spill between phrases. A gate closes when the track goes quiet. | **Expander/Gate** |
| **Sibilance** | Sharp S and T sounds on a voice. A de-esser turns down only those moments. | **De-Esser** |
| **Punch** | Drums feel soft, or too spiky. A transient shaper changes the hit without changing the tail. | **Transient** |
| **Space** | The part sounds dry and stuck to the speaker. Reverb puts it in a room. Delay adds echoes. | **Plate Reverb, Room Reverb, Reverb, ChamberMade, Delay, Mod Delay** |
| **Movement** | A static sound that needs width or shimmer. These effects make it shift slowly over time. | **Chorus, Flanger, Phaser, Mod** |
| **Colour** | Too clean. Saturation adds warmth and grit, from a touch of tape to full distortion. | **Saturator, Sat, Tape, Dist, Amp, Lo-Fi** |
| **Pitch** | A part that needs to be higher or lower, or a voice that needs tuning. | **Pitch Shifter, Pitch, and AI › Tune Vocal** |
| **Loudness** | The finished mix is too quiet. A maximizer raises the level and stops the peaks from clipping. | **Maximizer, LevelUp** |

**Instruments**

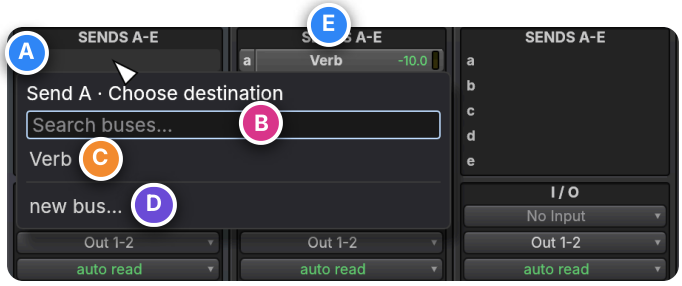
FreeDAWm also has instruments to play from MIDI: a synth, a drum synth, the Lift sampler and the Juke drum machine. Pick one when you make an Instrument Track.

FreeDAWm

# SENDS & SHARED REVERB

**The Goal**

Put several tracks in the same room. One reverb on an aux track, fed by a send from each track, sounds more natural than a separate reverb on every track and is lighter on the computer.




**A — Send slots**
Each strip has slots a to e. Click an empty one to open this menu.

**B — Search**
Type to find a bus when you have many.

**C — A bus**
Choose where the copy goes. Verb is the demo song's reverb bus.

**D — New bus**
Makes a new path if the one you want does not exist yet.

**E — Send level**
How much of this track goes to the bus. The Snare sends at −10 dB.

## BUILD ONE FROM NOTHING

**1 — Make an aux track**
Track › New, and choose Aux Input for the track type. Name it after its job, such as Verb.

**2 — Give it a bus**
In the aux's I/O section, set its input to a bus. Make a new one if you need to.

**3 — Add the reverb**
Click the plus at the top of the aux strip and choose a reverb. Set it fully wet, since the dry sound comes from the tracks themselves.

**4 — Send to it**
On each track, click a send slot and choose the same bus. Raise the send level until you can hear the room.

**Good to know**

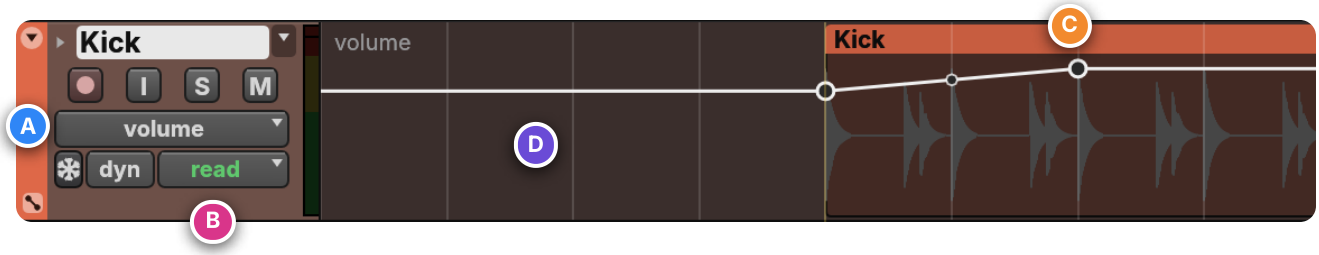
A send follows the track's fader. Turn a track down and less of it reaches the reverb, so the balance between dry and wet stays the same.

FreeDAWm

# AUTOMATION

**The Goal**

Make the mix change over time: a vocal that comes up in the chorus, a pad that fades in. Automation is a line you draw on the track, and FreeDAWm moves the control along it.




**A** **Track view**
Change waveform to volume, pan or mute to see that control's line over the clips.

**B** **Automation mode**
Decides whether the track follows the line, ignores it or records your moves.

**C** **Breakpoints**
The line bends at each point. Drag a point to move it.

**D** **The line**
Higher is louder. Here the kick comes in quiet and rises over two bars.

## TWO WAYS TO WRITE IT

**Draw it.** Show the volume view, then double-click the line with the Grabber or the Smart Tool to add a point. Use the Pencil to draw a shape freehand.

**Perform it.** Set the mode to touch, press play and move the fader. FreeDAWm records the move and returns to the old line when you let go.

| | |
|---|---|
| **off** | Ignores the line. The fader stays wherever you put it. |
| **read** | Follows the line. This is the mode tracks start in. |
| **touch** | Records while you hold a control, then goes back to the line. |
| **latch** | Records from the moment you touch a control, and holds your last value until you stop. |
| **write** | Records over everything while the song plays. Switch back to read when you finish. |
| **trim** | Raises or lowers the whole existing line without redrawing its shape. |

SECTION 5

# Finishing

Turn the session into a file people can play, send the separate parts to someone else, and take your work to another program.

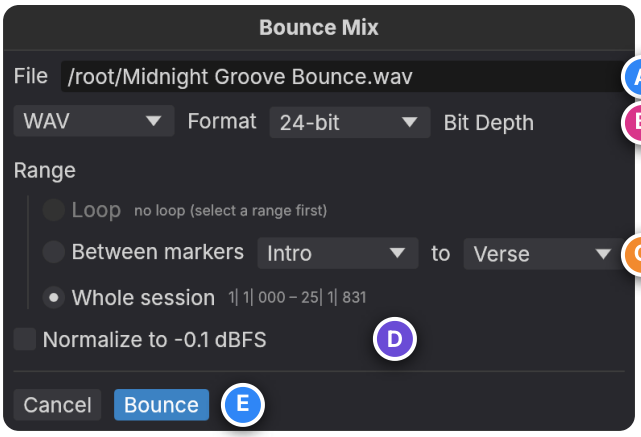
FreeDAWm

FreeDAWm

# BOUNCE THE MIX

**The Goal**

Make one audio file of the whole mix, the file you share or upload. Choose File › Bounce Mix, or press Cmd Alt B.




**A** **File**
Where the file is saved and what it is called.

**B** **Format and bit depth**
The kind of file and its quality. See the table below.

**C** **Range**
The loop, the stretch between two markers, or the whole session.

**D** **Normalize**
Raises the finished file so its loudest moment sits just under the limit.

**E** **Bounce**
Writes the file.

## WHICH FORMAT?

| | |
|---|---|
| **WAV or AIFF** | Full quality, large files. Use these for mastering, for distribution services, and to keep as your own copy. |
| **FLAC** | Full quality at about half the size. Good for archiving and sending. |
| **MP3 or Opus** | Small files that lose a little quality. Use these for email, messages and quick listens. |

## WHICH BIT DEPTH?

| | |
|---|---|
| **16-bit** | The CD standard. Fine for a finished song. |
| **24-bit** | The usual choice, and what to send to a mastering engineer. |
| **32-bit float** | Cannot clip. Use it for a mix you will keep working on elsewhere. |

**Before you bounce**

Listen to the whole song once and watch the master meter. If it reaches the red, pull the master fader down and bounce again.

FreeDAWm

# OTHER WAYS OUT

**The Goal**

Get more than one stereo file out of a session: the separate parts for a collaborator, the notes for another program, or the whole project.

| YOU WANT | WHAT YOU GET | WHERE |
|---|---|---|
| **Stems** | One audio file per track, all the same length, so someone else can remix or master from the parts. | **File › Export › Stems** |
| **Clips** | The selected clips as separate audio files, for samples or sound effects. | **File › Export › Clips** |
| **The notes** | A MIDI file of your MIDI and instrument parts, to open in any other music program. | **File › Export › MIDI** |
| **The project** | A DAWproject file, an open format that other workstations can read, with tracks and clips in place. | **File › Export › DAWproject** |
| **Some tracks** | A new session holding only the tracks you selected. | **File › Export › Selected Tracks as New Session** |
| **The words** | The lyrics as plain text, or as an LRC file with timings for players that show lyrics. | **File › Export › Lyrics** |
| **One track** | A track rendered to audio with its effects, back in the session. | **Track › Bounce or Track › Commit** |
| **A backup** | A full copy of the session in another place. | **File › Save Session Copy In** |

**Coming in**

The same menu works the other way. File › Import brings in audio, MIDI, video, DAWproject files and the tracks of another session.

# GLOSSARY

**Aux track**
A track with no clips of its own. It takes sound from a bus, usually to hold an effect that several tracks share.

**Automation**
A line on a track that moves a control, such as volume, as the song plays.

**Bounce**
To write the mix, or one track, out as an audio file.

**Buffer size**
How much sound the computer handles in one go. Small means less delay, large means steadier playback.

**Bus**
A path inside FreeDAWm that carries sound from one or more tracks to another place.

**Clip**
One piece of audio or MIDI on the timeline. Editing a clip does not change the recording behind it.

**Comp**
One good take assembled from the best parts of several.

**Compressor**
An effect that turns down the loud moments so a part sits at a steadier level.

**Count-off**
Bars of click played before recording starts, to give you the tempo.

**Crossfade**
One clip fading out while the next fades in, to hide the join.

**DAW**
Digital audio workstation: a program for recording, editing and mixing music.

**dB**
Decibel, the unit of level. A fader at 0 dB leaves the level unchanged.

**EQ**
Equaliser. It turns ranges of a sound up or down, from bass to treble.

**Fade**
A ramp in level at the start or end of a clip.

**Fader**
The slider that sets a track's level in the mix.

**Freeze**
To swap a track's effects for a rendered copy, so they stop using the computer.

**Grid**
The steps that clips and selections snap to in Grid mode, such as a sixteenth note.

**Input monitoring**
Hearing a track's live input through FreeDAWm while you play.

**Insert**
An effect placed directly on a track.

**Latency**
The delay between playing a note and hearing it.

**Marker**
A named point on the timeline, such as Verse or Chorus.

**Master**
The last fader in the mixer. Everything passes through it.

# GLOSSARY

**MIDI**
Notes and performance data, not sound. An instrument turns it into sound.

**Mono and stereo**
Mono is one channel. Stereo is two, left and right.

**Nudge**
To move a clip or selection by a small fixed step.

**Pan**
A track's position between the left and right speakers.

**Playlist**
One version of a track's clips. A track can hold several, such as one per take.

**Plugin**
An effect or instrument, built in or installed separately.

**Pre-roll**
Playback that starts a little before the point you want, to lead you in.

**Punch in**
To record over one part of a take and leave the rest.

**Quantize**
To pull notes or clips onto the grid.

**Record enable**
To arm a track so that pressing Record captures to it.

**Reverb**
An effect that adds the sound of a room or hall.

**Sample rate**
How many times a second the sound is measured. 48 000 Hz is common.

**Send**
A copy of a track's sound routed to a bus, with its own level.

**Session**
A FreeDAWm project: its tracks, clips, mix and settings.

**Solo**
To hear one track by silencing the others.

**Stem**
One part of a mix exported as its own file.

**Take**
One recorded attempt at a part.

**Tempo**
The speed of the song, in beats per minute.

**Track**
One lane in the session, with its own clips and its own strip in the mixer.

**Transient**
The sharp start of a sound, such as a drum hit.

**Transport**
The play, stop and record controls.

**Velocity**
How hard a MIDI note is played.